MILITARY SPECIFICATION

RESISTOR, FIXED, COMPOSITION (INSULATED), STYLE RC05

INACTIVE FOR DESIGN AFTER 11 AUGUST 1969 (SEE PAGE 3). USE MIL-R-39008/4.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RC05 resistor.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-11 - Resistors, Fixed, Composition (Insulated), General Specification For.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-11, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/8 watt, based on full load operation at an ambient temperature of 70° C. Resistors operated at ambient temperatures in excess of 70° C shall be derated in accordance with figure 2. The rated direct current (dc) or root mean square (rms) continuous working voltage shall not exceed 150 volts.

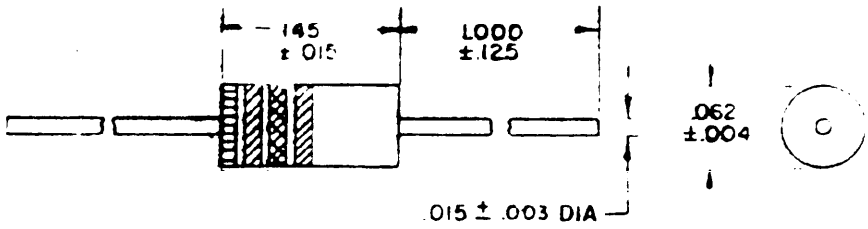
3.4 Resistance. Minimum resistance value shall be 10 ohms; maximum resistance value shall be 22 megohms.

3.5 Moisture resistance. The change in resistance shall not exceed an average of 12 percent for each group of 10 resistors tested, nor a maximum of 15 percent for any individual resistor up to and including 1 megohm. For values over 1 megohm, the change in resistance shall not exceed a maximum of 15 percent for any individual resistor.

3.6 Life. The change in resistance between the initial measurement and each of the succeeding measurements shall not exceed an average of 6 percent for each group of 10 resistors tested, nor a maximum of 8 percent for any individual resistor.

3.7 Resistance to soldering heat. The change in resistance shall not exceed ± (3 percent + 0.05 ohm).

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INCHES	MM	INCHES	MM
.003	.08	.062	1.57
.004	.10	.125	3.18
.015	.38	.145	3.68
		1.000	25.40

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style RC05 resistor.

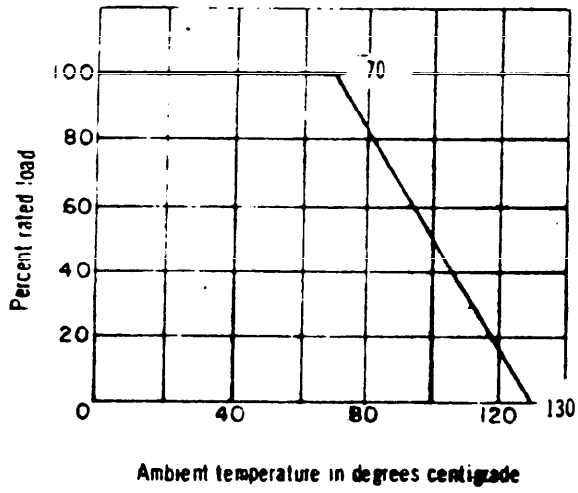


FIGURE 2. Derating curve for high ambient temperatures.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-11, and as specified herein.

4.2 Dielectric withstanding voltage. Test voltage at atmospheric pressure shall be 300 volts.

4.3 Terminal strength.

4.3.1 Direct load. Direct load shall be applied gradually until the load reaches 2 pounds.

4.3.2 Twist. The twist test is applicable to this specification.

4.4 Resistance to soldering heat. Temperature of solder in which resistor terminals are immersed shall be 250 ±10° C.

4.5 Seal. The seal test is not applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-11.

6. NOTES

6.1 The notes specified in MIL-R-11 are applicable to this specification.

6.2 Style supersession. Style RC05 in this specification supersedes style RC06 of MIL-R-11/10A dated 17 May 1963.

* 6.3 Substitution data. The resistors specified herein are not for use in design after 11 August 1969. They are suitable for use in design contracts current to that date and to support existing military equipment. Resistors specified in MIL-R-39008/4 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value and tolerance, of this specification.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0727)

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-R-11/11E, RESISTOR, FIXED, COMPOSITION (INSULATED), STYLE RC05

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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